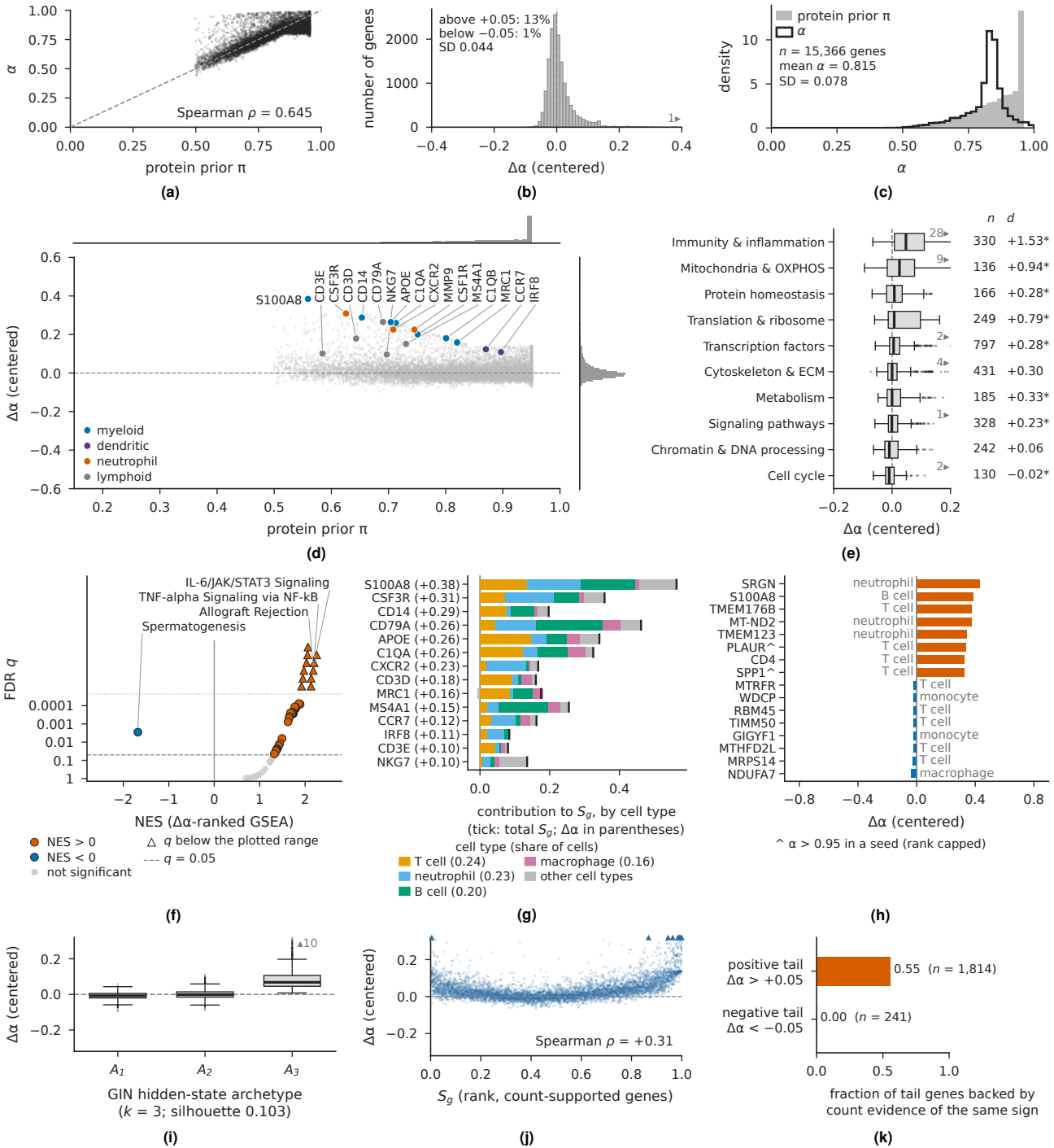

Spearman ρ = 0.645
protein prior π
α
(a)
above +0.05: 13%
below −0.05: 1%
SD 0.044
number of genes
Δα (centered)
(b)
protein prior π
α
n = 15,366 genes
mean α = 0.815
SD = 0.078
density
(c)
S100A8
CD3E
CSF3R
CD3D
CD14
CD79A
NKG7
APOE
C1QA
CXCR2
MMP9
CSF1R
MS4A1
C1QB
MRC1
CCR7
IRF8
myeloid
dendritic
neutrophil
lymphoid
Δα (centered)
protein prior π
(d)
n
d
Immunity & inflammation 330 +1.53*
Mitochondria & OXPHOS 136 +0.94*
Protein homeostasis 166 +0.28*
Translation & ribosome 249 +0.79*
Transcription factors 797 +0.28*
Cytoskeleton & ECM 431 +0.30
Metabolism 185 +0.33*
Signaling pathways 328 +0.23*
Chromatin & DNA processing 242 +0.06
Cell cycle 130 −0.02*
Δα (centered)
(e)
IL-6/JAK/STAT3 Signaling
TNF-alpha Signaling via NF-kB
Allograft Rejection
Spermatogenesis
FDR q
NES (Δα-ranked GSEA)
NES > 0
NES < 0
not significant
△ q below the plotted range
q = 0.05
(f)
S100A8 (+0.38)
CSF3R (+0.31)
CD14 (+0.29)
CD79A (+0.26)
APOE (+0.26)
C1QA (+0.26)
CXCR2 (+0.23)
CD3D (+0.18)
MRC1 (+0.16)
MS4A1 (+0.15)
CCR7 (+0.12)
IRF8 (+0.11)
CD3E (+0.10)
NKG7 (+0.10)
contribution to S_g, by cell type
(tick: total S_g; Δα in parentheses)
cell type (share of cells)
T cell (0.24)
macrophage (0.16)
neutrophil (0.23)
other cell types
B cell (0.20)
(g)
SRGN neutrophil
S100A8 B cell
TMEM176B T cell
MT-ND2 neutrophil
TMEM123 neutrophil
PLAUR^ T cell
CD4 T cell
SPP1^ T cell
MTRFR T cell
WDCP monocyte
RBM45 T cell
TIMM50 T cell
GIGYF1 monocyte
MTHFD2L T cell
MRPS14 T cell
NDUFA7 macrophage
Δα (centered)
^ α > 0.95 in a seed (rank capped)
(h)
Δα (centered)
A_1
A_2
A_3
GIN hidden-state archetype
(k = 3; silhouette 0.103)
(i)
Spearman ρ = +0.31
S_g (rank, count-supported genes)
(j)
positive tail
Δα > +0.05
0.55 (n = 1,814)
negative tail
Δα < −0.05
0.00 (n = 241)
fraction of tail genes backed by
count evidence of the same sign
(k)